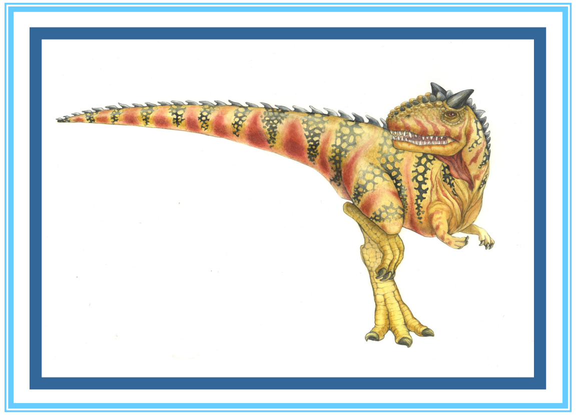
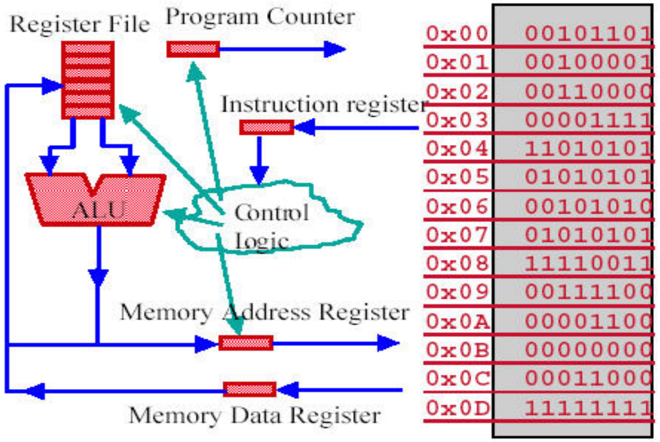
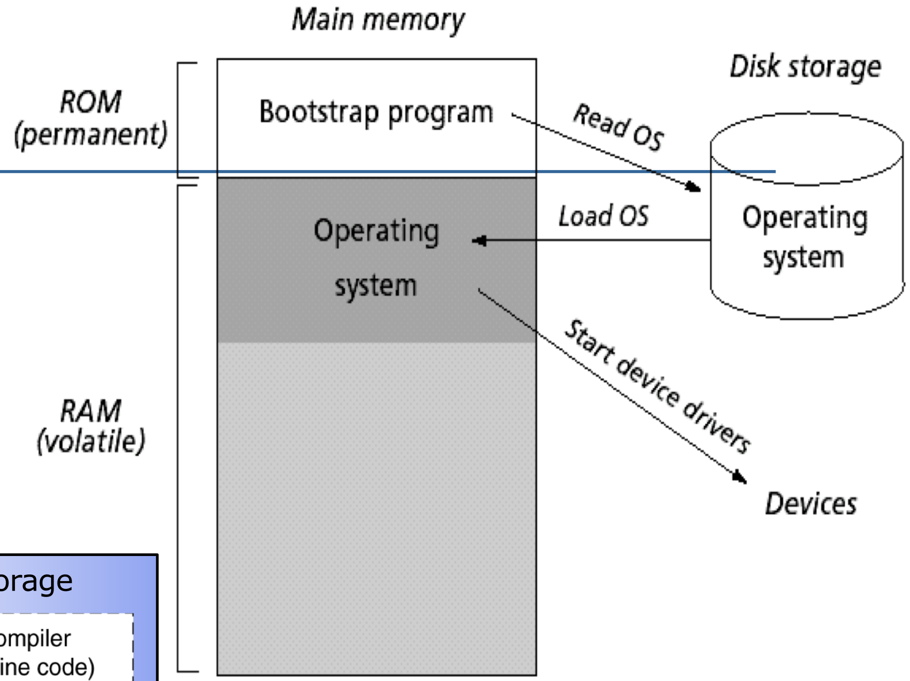
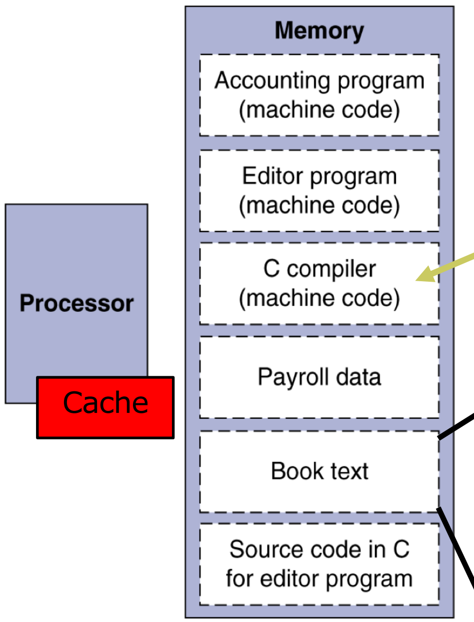
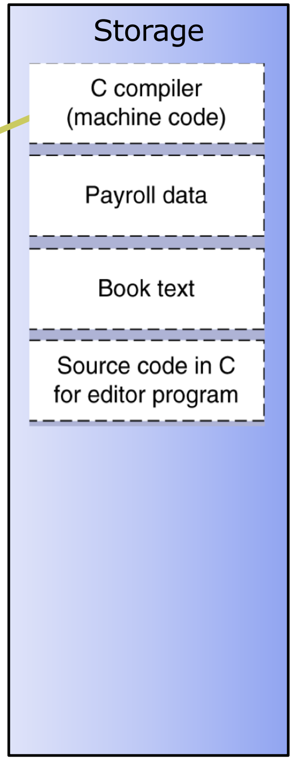
# Chapter 6: Process Synchronization

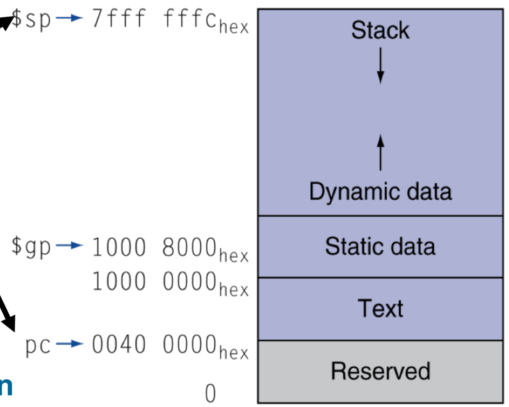


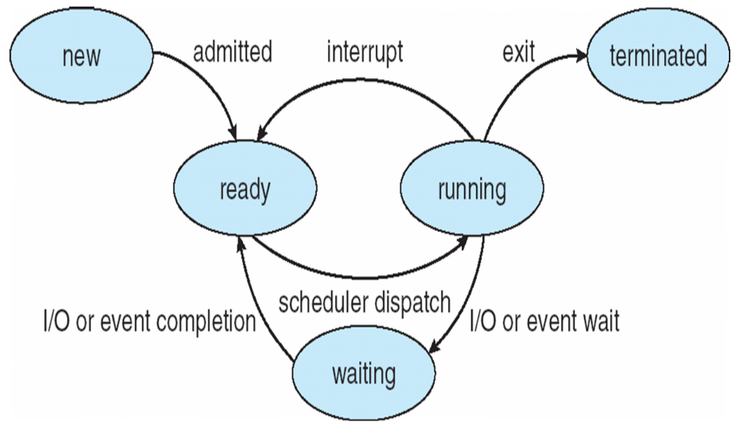

Register File
Program Counter
Instruction register
ALU
Control logic
Memory Address Register
Memory Data Register
0x00 00101101
0x01 00100001
0x02 00110000
0x03 00001111
0x04 11010101
0x05 01010101
0x06 00101010
0x07 01010101
0x08 11110011
0x09 00111100
0x0A 00001100
0x0B 00000000
0x0C 00011000
0x0D 11111111

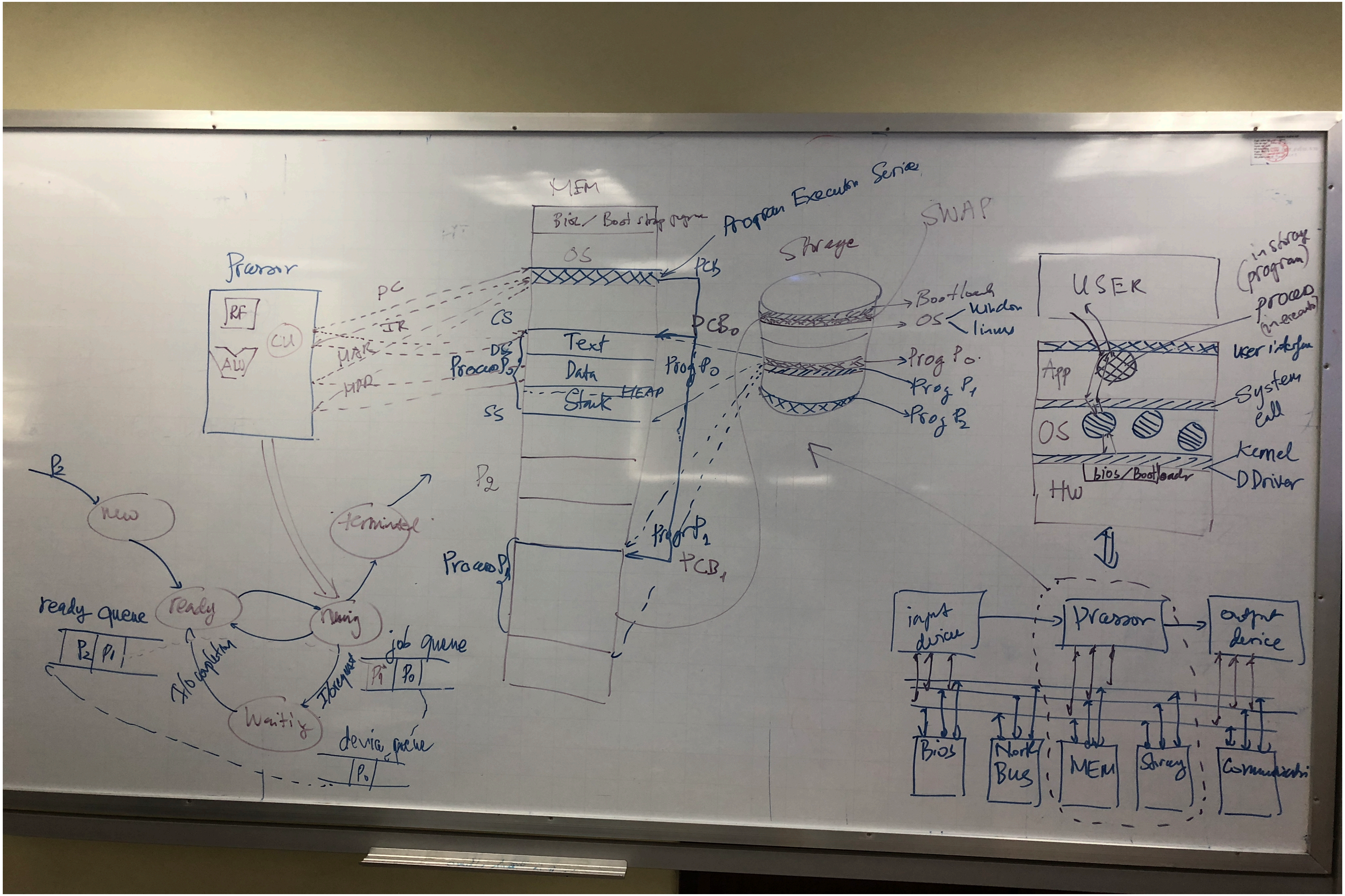

Main memory
Disk storage
ROM (permanent)
Bootstrap program
Read OS
Operating system
Load OS
Operating system
RAM (volatile)
Start device drivers
Devices


Processor
Cache
Memory
Accounting program (machine code)
Editor program (machine code)
C compiler (machine code)
Payroll data
Book text
Source code in C for editor program


Storage
C compiler (machine code)
Payroll data
Book text
Source code in C for editor program


$sp → 7fff fffc hex
$gp → 1000 8000 hex
1000 0000 hex
pc → 0040 0000 hex
0
Stack
Dynamic data
Static data
Text
Reserved

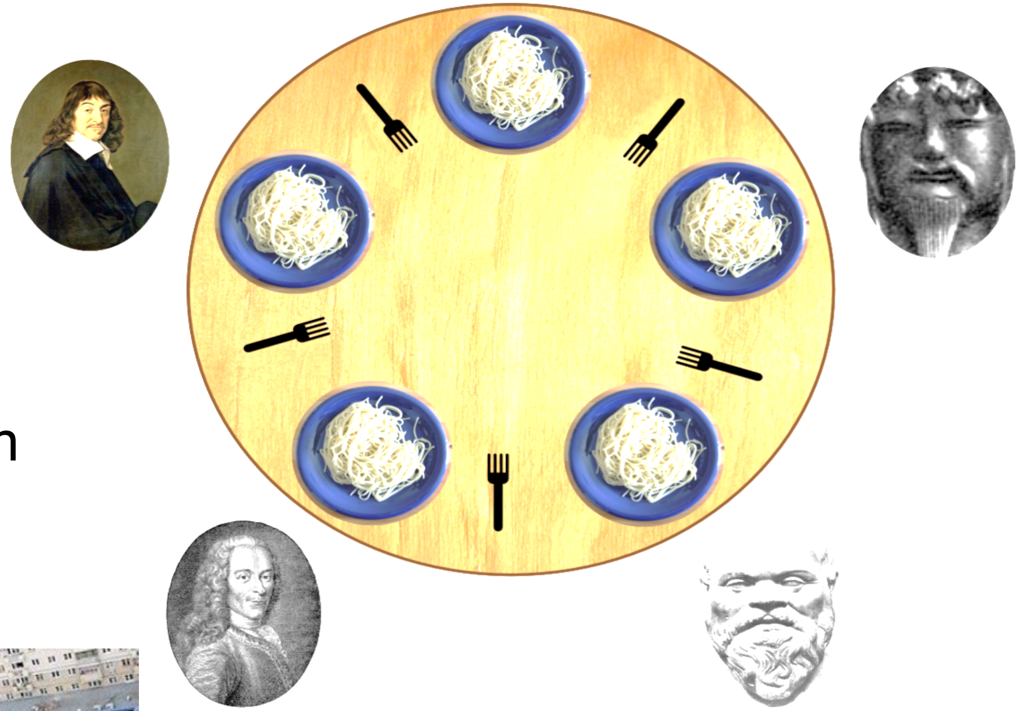

new
admitted
interrupt
exit
terminated
ready
running
I/O or event completion
scheduler dispatch
I/O or event wait
waiting



MEM
Bios / Boot strap program
OS
PCB
Program Execution Series
Storage
SWAP
Processor
RF
CU
ALU
PC
IR
MAR
MDR
CS
DS
SS
Process P0
Text
Data
Stack
HEAP
PCB0
Prog P0
P2
Prog P1
PCB1
Process P1
Bootloader
OS
Window
linux
Prog P0
Prog P1
Prog P2
USER
App
OS
Hw
bios/Bootloader
in storage (program)
process (in execution)
User interface
System call
Kernel
D Driver
new
ready
terminated
waiting
ready queue
P2
P1
job queue
P1
P0
I/O request
I/O completion
device queue
P0
input device
Processor
output device
Bios
North Bus
MEM
Storage

# Module 6: Process Synchronization

- Background
- The Critical-Section Problem
- Peterson's Solution
- Synchronization Hardware
- Semaphores
- Classic Problems of Synchronization
- Monitors
- Synchronization Examples

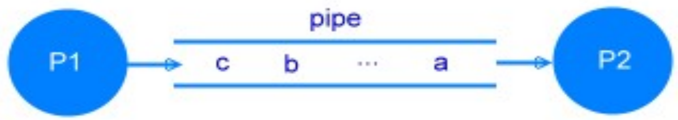


# Objectives

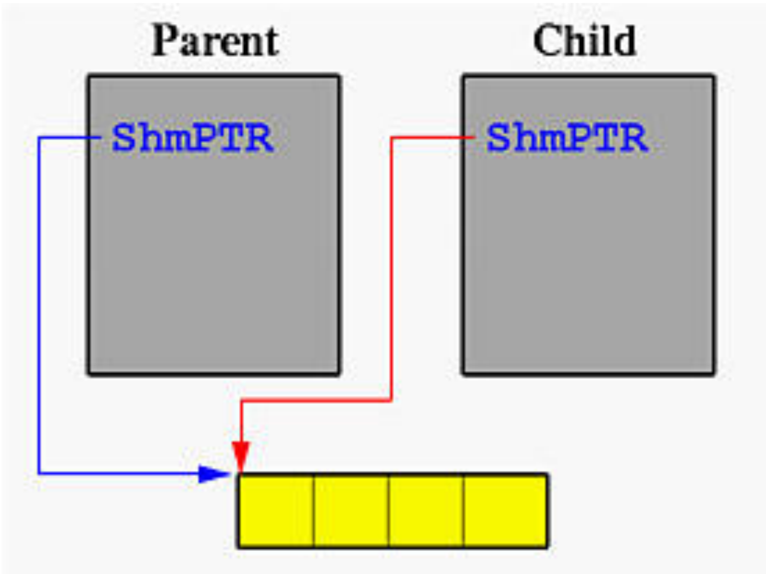
- To introduce the critical-section problem, whose solutions can be used to ensure the consistency of shared data
- To present both software and hardware solutions of the critical-section problem
- To introduce the concept of an atomic transaction and describe mechanisms to ensure atomicity

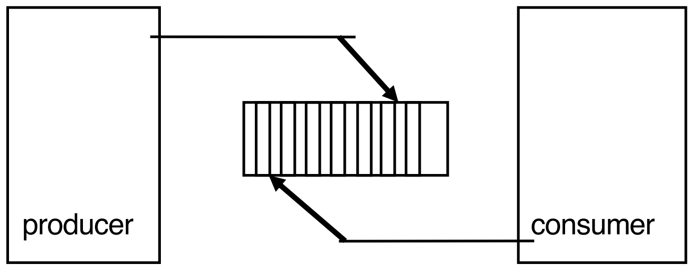
| *time* | Person A | Person B |
|---|---|---|
| 8:00 | Look in fridge. *Out of milk* | |
| 8:05 | Leave for store. | |
| 8:10 | Arrive at store. | Look in fridge. *Out of milk* |
| 8:15 | Buy milk. | Leave for store. |
| 8:20 | Leave the store. | Arrive at store. |
| 8:25 | Arrive home, put milk away. | Buy milk. |
| 8;30 | | Leave the store. |
| 8:35 | | Arrive home, *OH! OH!* |

# Background




- Concurrent access to shared data may result in data inconsistency
- Maintaining data consistency requires mechanisms to ensure the orderly execution of cooperating processes
- Suppose that we wanted to provide a solution to the consumer-producer problem that fills **all** the buffers. We can do so by having an integer **count** that keeps track of the number of full buffers. Initially, count is set to 0. It is incremented by the producer after it produces a new buffer and is decremented by the consumer after it consumes a buffer.



# Producer

```
#define BUFFER_SIZE 10
typedef struct {
    DATA          data;
} item;
item  buffer[BUFFER_SIZE];
int   in = 0;     // Location of next input to buffer
int   out = 0;    // Location of next removal from buffer
int     counter = 0;  // Number of buffers currently full
```

```
while (true) {                                   // PRODUCER //


      /*  produce an item and put in nextProduced  */
       while (counter == BUFFER_SIZE)
             ; // do nothing
          buffer [in] = nextProduced;
          in = (in + 1) % BUFFER_SIZE;
          counter++;
}
```

# Consumer

```
#define BUFFER_SIZE 10
typedef struct {
    DATA          data;
} item;
item  buffer[BUFFER_SIZE];
int   in = 0;              // Location of next input to buffer
int   out = 0;             // Location of next removal from buffer
int    counter = 0;        // Number of buffers currently full
```

```
while (true)  {                                  // CUNSUMER //
        while (counter == 0)
            ; // do nothing
            nextConsumed =  buffer[out];
             out = (out + 1) % BUFFER_SIZE;
                counter--;

              /*  consume the item in nextConsumed  */
 }
```

# Race Condition

- counter++ could be implemented as

  register1 = counter
  register1 = register1 + 1
  counter = register1

- counter-- could be implemented as

  register2 = counter
  register2 = register2 - 1
  count = register2

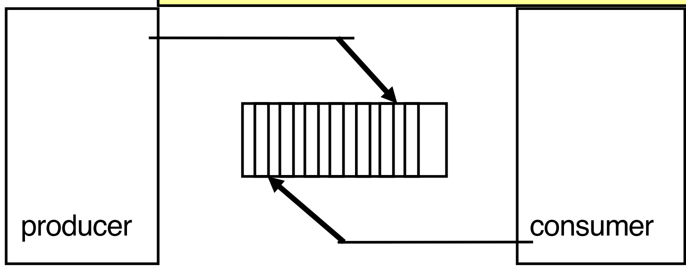
```
#define BUFFER_SIZE 10
typedef struct {
    DATA      data;
} item;
item   buffer[BUFFER_SIZE];
int    in = 0;
int    out = 0;
int      counter = 0;
```

**PRODUCER**

```
item   nextProduced;

while (TRUE) {
    while (counter ==
     BUFFER_SIZE);
    buffer[in] = nextProduced;
    in = (in + 1) % BUFFER_SIZE;
    counter++;
}
```

**CONSUMER**

```
item  nextConsumed;

while (TRUE) {
    while (counter == 0);
    nextConsumed = buffer[out];
    out = (out + 1) %
     BUFFER_SIZE;
    counter--;
}
```

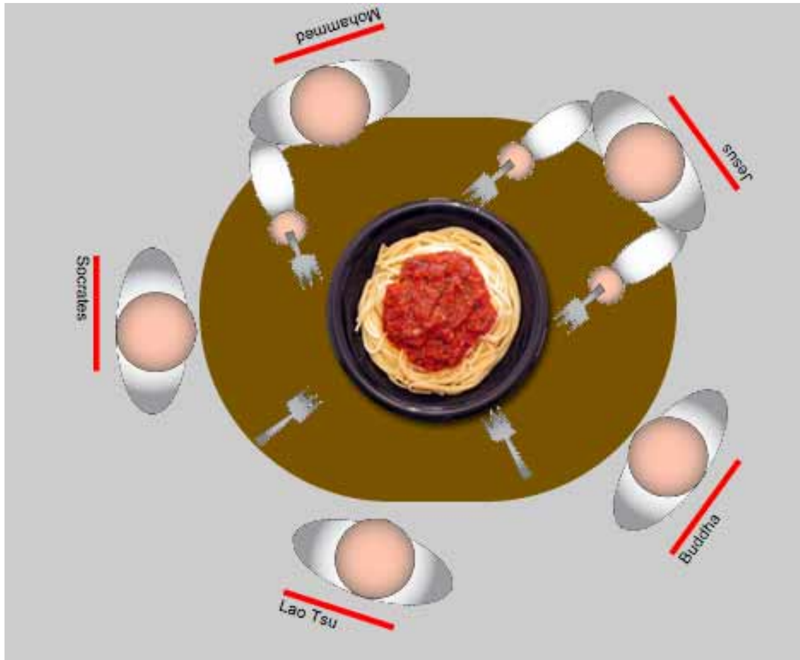



- Consider this execution interleaving with "count = 5" initially:

  S0: producer execute register1 = counter {register1 = 5}
  S1: producer execute register1 = register1 + 1 {register1 = 6}
  S2: consumer execute register2 = counter {register2 = 5}
  S3: consumer execute register2 = register2 - 1 {register2 = 4}
  S4: producer execute counter = register1 {count = 6 }
  S5: consumer execute counter = register2 {count = 4}

# Critical Section Problem

- Consider system of n processes $\{p_0, p_1, \ldots p_{n-1}\}$
- Each process has **critical section** segment of code
  - Process may be changing common variables, updating table, writing file, etc
  - When one process in critical section, no other may be in its critical section
- Critical section problem is to design protocol to solve this
- Each process must ask permission to enter critical section in **entry section**, may follow critical section with **exit section**, then **remainder section**
- Especially challenging with preemptive kernels

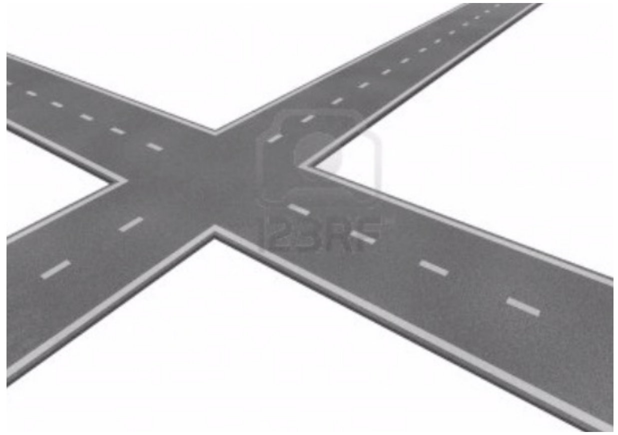


# Critical Section

- General structure of process $p_i$ is

```
do {
        entry section
                critical section
        exit section
                remainder section
} while (TRUE);
```

**Figure 6.1** General structure of a typical process $P_i$.



# Solution to Critical-Section Problem

1. **Mutual Exclusion** - If process $P_i$ is executing in its critical section, then no other processes can be executing in their critical sections

2. **Progress** - If no process is executing in its critical section and there exist some processes that wish to enter their critical section, then the selection of the processes that will enter the critical section next cannot be postponed indefinitely

3. **Bounded Waiting** - A bound must exist on the number of times that other processes are allowed to enter their critical sections after a process has made a request to enter its critical section and before that request is granted
   - Assume that each process executes at a nonzero speed
   - No assumption concerning **relative speed** of the n processes

# Peterson's Solution

- Two process solution
- Assume that the LOAD and STORE instructions are atomic; that is, cannot be interrupted
- The two processes share two variables:
  - int **turn**;
  - Boolean **flag[2]**
- The variable **turn** indicates whose turn it is to enter the critical section
- The **flag** array is used to indicate if a process is ready to enter the critical section. **flag[i]** = true implies that process $P_i$ is ready!

# Algorithm for Process $P_i$

| Pi | PJ |
|---|---|
| <pre>do {<br>        flag[i] = TRUE;<br>        turn = j;<br>        while (flag[j] && turn == j);<br>                critical section<br>        flag[i] = FALSE;<br>                remainder section<br>    } while (TRUE);</pre> | <pre>do {<br>        flag[j] = TRUE;<br>        turn = i;<br>        while (flag[i] && turn == i);<br>                critical section<br>        flag[j] = FALSE;<br>                remainder section<br>    } while (TRUE);</pre> |

- Provable that

1. Mutual exclusion is preserved
2. Progress requirement is satisfied
3. Bounded-waiting requirement is met

# Synchronization Hardware

- Many systems provide hardware support for critical section code
- Uniprocessors – could disable interrupts
  - Currently running code would execute without preemption
  - Generally too inefficient on multiprocessor systems
    - Operating systems using this not broadly scalable
- Modern machines provide special atomic hardware instructions
    - Atomic = non-interruptable
  - Either test memory word and set value
  - Or swap contents of two memory words

# Solution to Critical-section Problem Using Locks

```
do {
    acquire lock
            critical section
    release lock
            remainder section
} while (TRUE);
```

# TestAndSet Instruction

- Definition:

```
boolean TestAndSet (boolean *target)
{
    boolean rv = *target;
    *target = TRUE;
    return rv:
}
```

# Solution using TestAndSet

- Shared boolean variable lock, initialized to FALSE
- Solution:

```
do {
        while ( TestAndSet (&lock ))
                ;   // do nothing

              //    critical section

        lock = FALSE;

              //      remainder section

} while (TRUE);
```

# Swap Instruction

- Definition:

```
void Swap (boolean *a, boolean *b)
{
        boolean temp = *a;
        *a = *b;
        *b = temp:
}
```

# Solution using Swap

- Shared Boolean variable lock initialized to FALSE; Each process has a local Boolean variable key
- Solution:

```
do {
        key = TRUE;
        while ( key == TRUE)
               Swap (&lock, &key );

                  //    critical section

         lock = FALSE;

                 //      remainder section

 } while (TRUE);
```

# Bounded-waiting Mutual Exclusion with TestandSet()

```
do {
      waiting[i] = TRUE;
      key = TRUE;
      while (waiting[i] && key)
                key = TestAndSet(&lock);
      waiting[i] = FALSE;
                // critical section
      j = (i + 1) % n;
      while ((j != i) && !waiting[j])
                j = (j + 1) % n;
      if (j == i)
                lock = FALSE;
      else
                waiting[j] = FALSE;
                // remainder section
} while (TRUE);
```

# Semaphore

- Synchronization tool that does not require busy waiting
- Semaphore *S* – integer variable
- Two standard operations modify S: wait() and signal()
  - Originally called P() and V()
- Less complicated
- Can only be accessed via two indivisible (atomic) operations
  - ```
    wait (S) {
         while S <= 0
                  ; // no-op
           S--;
     }
    ```
  - ```
    signal (S) {
       S++;
    }
    ```

# Semaphore as General Synchronization Tool

- **Counting** semaphore – integer value can range over an unrestricted domain
- **Binary** semaphore – integer value can range only between 0 and 1; can be simpler to implement
  - Also known as **mutex locks**
- Can implement a counting semaphore S as a binary semaphore
- Provides mutual exclusion

```
Semaphore mutex;    //  initialized to 1
do {
    wait (mutex);
        // Critical Section
    signal (mutex);
        // remainder section
} while (TRUE);
```

# Semaphore Implementation

- Must guarantee that no two processes can execute wait () and signal () on the same semaphore at the same time
- Thus, implementation becomes the critical section problem where the wait and signal code are placed in the critical section
  - Could now have **busy waiting** in critical section implementation
    - But implementation code is short
    - Little busy waiting if critical section rarely occupied
- Note that applications may spend lots of time in critical sections and therefore this is not a good solution

# Semaphore Implementation with no Busy waiting

- With each semaphore there is an associated waiting queue
- Each entry in a waiting queue has two data items:
  - value (of type integer)
  - pointer to next record in the list
- Two operations:
  - **block** – place the process invoking the operation on the appropriate waiting queue
  - **wakeup** – remove one of processes in the waiting queue and place it in the ready queue

# Semaphore Implementation with no Busy waiting (Cont.)

- Implementation of wait:

```
wait(semaphore *S) {
        S->value--;
        if (S->value < 0) {
                add this process to S->list;
                block();
        }
}
```

- Implementation of signal:

```
signal(semaphore *S) {
        S->value++;
        if (S->value <= 0) {
                remove a process P from S->list;
                wakeup(P);
        }
}
```

# Deadlock and Starvation

- **Deadlock** – two or or more processes are waiting indefinitely for an event that can be caused by only one of the waiting processes
- Let S and Q be two semaphores initialized to 1

| $P_0$ | $P_1$ |
|---|---|
| wait (S); | wait (Q); |
| wait (Q); | wait (S); |
| . | . |
| . | . |
| . | . |
| signal (S); | signal (Q); |
| signal (Q); | signal (S); |

- **Starvation** – indefinite blocking
  - A process may never be removed from the semaphore queue in which it is suspended
- **Priority Inversion** – Scheduling problem when lower-priority process holds a lock needed by higher-priority process
  - Solved via **priority-inheritance protocol**

# Classical Problems of Synchronization

- Classical problems used to test newly-proposed synchronization schemes
  - Bounded-Buffer Problem
  - Readers and Writers Problem
  - Dining-Philosophers Problem

# Bounded-Buffer Problem

- *N* buffers, each can hold one item
- Semaphore mutex initialized to the value 1
- Semaphore full initialized to the value 0
- Semaphore empty initialized to the value N

# Bounded Buffer Problem (Cont.)

- The structure of the producer process

```
do  {

            //   produce an item in nextp

        wait (empty);
        wait (mutex);

            //  add the item to the  buffer

        signal (mutex);
        signal (full);
   } while (TRUE);
```

# Bounded Buffer Problem (Cont.)

- The structure of the consumer process

```
do {
        wait (full);
        wait (mutex);

              //  remove an item from  buffer to nextc

        signal (mutex);
        signal (empty);

              //  consume the item in nextc

} while (TRUE);
```

# Readers-Writers Problem

- A data set is shared among a number of concurrent processes
  - Readers – only read the data set; they do **not** perform any updates
  - Writers – can both read and write
- Problem – allow multiple readers to read at the same time
  - Only one single writer can access the shared data at the same time
- Several variations of how readers and writers are treated – all involve priorities
- Shared Data
  - Data set
  - Semaphore mutex initialized to 1
  - Semaphore wrt initialized to 1
  - Integer readcount initialized to 0

# Readers-Writers Problem (Cont.)

- The structure of a writer process

```
do {
        wait (wrt) ;

            //    writing is performed

        signal (wrt) ;
} while (TRUE);
```

# Readers-Writers Problem (Cont.)

- The structure of a reader process

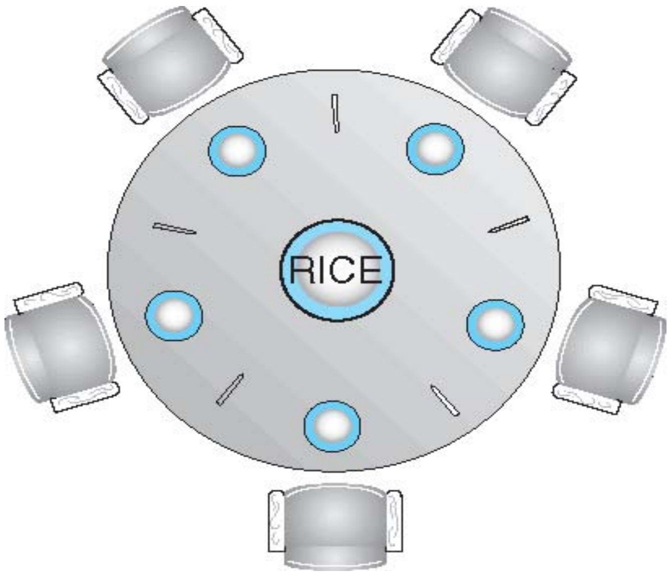
```
do {
        wait (mutex) ;
        readcount ++ ;
        if (readcount == 1)
                wait (wrt) ;
        signal (mutex)

                // reading is performed

        wait (mutex) ;
        readcount  - - ;
        if (readcount  == 0)
                signal (wrt) ;
        signal (mutex) ;
    } while (TRUE);
```

# Readers-Writers Problem Variations

- *First* variation – no reader kept waiting unless writer has permission to use shared object
- *Second* variation – once writer is ready, it performs write asap
- Both may have starvation leading to even more variations
- Problem is solved on some systems by kernel providing reader-writer locks

# Dining-Philosophers Problem

- Philosophers spend their lives thinking and eating
- Don't interact with their neighbors, occasionally try to pick up 2 chopsticks (one at a time) to eat from bowl
  - Need both to eat, then release both when done
- In the case of 5 philosophers
  - Shared data
    - Bowl of rice (data set)
    - Semaphore chopstick [5] initialized to 1



# Dining-Philosophers Problem Algorithm

- The structure of Philosopher *i*:

```
do  {
        wait ( chopstick[i] );
        wait ( chopStick[ (i + 1) % 5] );

            //  eat

        signal ( chopstick[i] );
        signal (chopstick[ (i + 1) % 5] );

            //  think

} while (TRUE);
```

- What is the problem with this algorithm?

# Problems with Semaphores

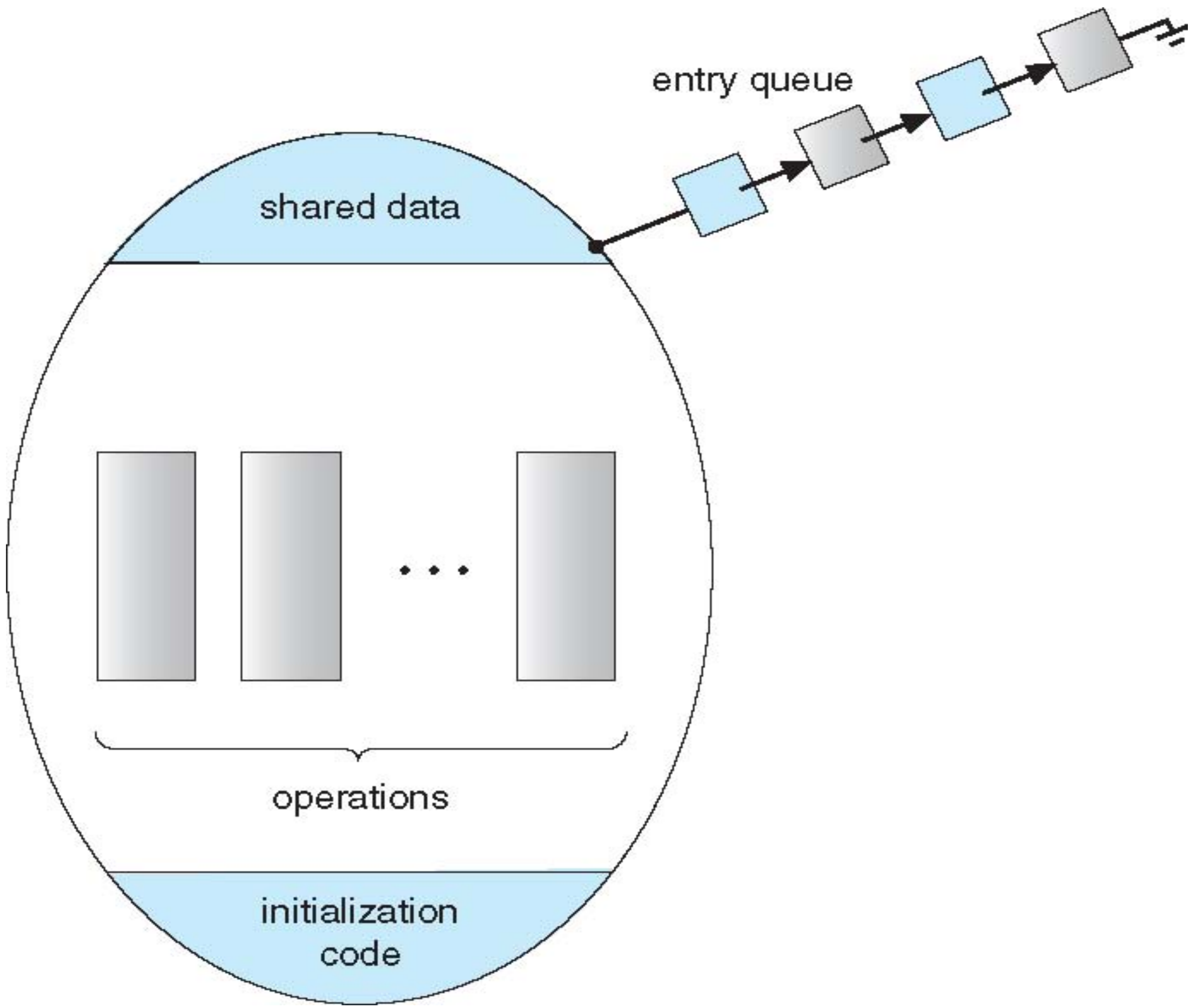
- Incorrect use of semaphore operations:
  - signal (mutex) …. wait (mutex)
  - wait (mutex) … wait (mutex)
  - Omitting of wait (mutex) or signal (mutex) (or both)
- Deadlock and starvation

# Monitors

- A high-level abstraction that provides a convenient and effective mechanism for process synchronization
- *Abstract data type*, internal variables only accessible by code within the procedure
- Only one process may be active within the monitor at a time
- But not powerful enough to model some synchronization schemes

```
monitor monitor-name
{
    // shared variable declarations
    procedure P1 (…) { …. }

    procedure Pn (…) {……}

    Initialization code (…) { … }
    }
}
```

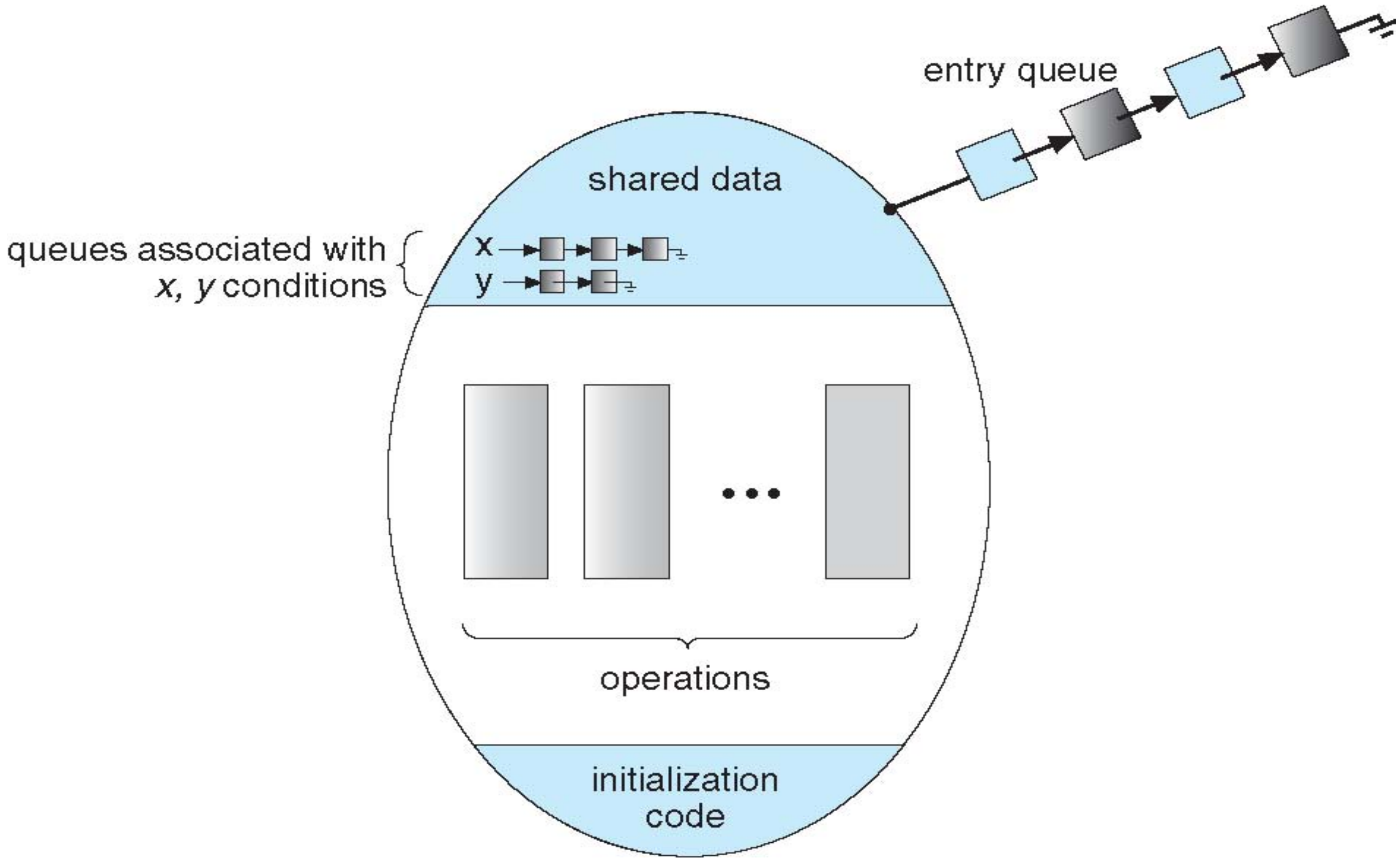
# Schematic view of a Monitor



# Condition Variables

- condition x, y;
- Two operations on a condition variable:
  - x.wait () – a process that invokes the operation is suspended until x.signal ()
  - x.signal () – resumes one of processes (if any) that invoked x.wait ()
    - If no x.wait () on the variable, then it has no effect on the variable

# Monitor with Condition Variables



# Condition Variables Choices

- If process P invokes x.signal (), with Q in x.wait () state, what should happen next?
  - If Q is resumed, then P must wait
- Options include
  - **Signal and wait** – P waits until Q leaves monitor or waits for another condition
  - **Signal and continue** – Q waits until P leaves the monitor or waits for another condition
  - Both have pros and cons – language implementer can decide
  - Monitors implemented in Concurrent Pascal compromise
    - P executing signal immediately leaves the monitor, Q is resumed
  - Implemented in other languages including Mesa, C#, Java

# Solution to Dining Philosophers

```
monitor DiningPhilosophers
  {
    enum { THINKING; HUNGRY, EATING) state [5] ;
    condition self [5];

    void pickup (int i) {
           state[i] = HUNGRY;
           test(i);
           if (state[i] != EATING) self [i].wait;
    }

     void putdown (int i) {
           state[i] = THINKING;
                 // test left and right neighbors
            test((i + 4) % 5);
            test((i + 1) % 5);
      }
```

# Solution to Dining Philosophers (Cont.)

```
void test (int i) {
        if ( (state[(i + 4) % 5] != EATING) &&
        (state[i] == HUNGRY) &&
        (state[(i + 1) % 5] != EATING) ) {
             state[i] = EATING ;
         self[i].signal () ;
         }
 }

 initialization_code() {
       for (int i = 0; i < 5; i++)
       state[i] = THINKING;
}
}
```

# Solution to Dining Philosophers (Cont.)

- Each philosopher *i* invokes the operations pickup() and putdown() in the following sequence:

  DiningPhilosophers.pickup (i);

  EAT

  DiningPhilosophers.putdown (i);

- No deadlock, but starvation is possible

# Monitor Implementation Using Semaphores

- Variables

```
semaphore mutex;  // (initially  = 1)
semaphore next;     // (initially  = 0)
int next_count = 0;
```

- Each procedure ***F*** will be replaced by

```
wait(mutex);
     …
               body of F;

     …
if (next_count > 0)
   signal(next)
else
   signal(mutex);
```

- Mutual exclusion within a monitor is ensured

# Monitor Implementation – Condition Variables

- For each condition variable ***x***, we have:

```
semaphore x_sem; // (initially  = 0)
int x_count = 0;
```

- The operation x.wait can be implemented as:

```
x-count++;
if (next_count > 0)
      signal(next);
else
      signal(mutex);
wait(x_sem);
x-count--;
```

# Monitor Implementation (Cont.)

- The operation x.signal can be implemented as:

```
if (x-count > 0) {
    next_count++;
    signal(x_sem);
    wait(next);
    next_count--;
}
```

# Resuming Processes within a Monitor

- If several processes queued on condition x, and x.signal() executed, which should be resumed?
- FCFS frequently not adequate
- **conditional-wait** construct of the form x.wait(c)
  - Where c is **priority number**
  - Process with lowest number (highest priority) is scheduled next

# A Monitor to Allocate Single Resource

```
monitor ResourceAllocator
{
      boolean busy;
      condition x;
      void acquire(int time) {
                  if (busy)
                        x.wait(time);
                  busy = TRUE;
      }
      void release() {
                  busy = FALSE;
                  x.signal();
      }
initialization code() {
       busy = FALSE;
      }
}
```

# Synchronization Examples

- Solaris
- Windows XP
- Linux
- Pthreads

# Solaris Synchronization

- Implements a variety of locks to support multitasking, multithreading (including real-time threads), and multiprocessing
- Uses **adaptive mutexes** for efficiency when protecting data from short code segments
  - Starts as a standard semaphore spin-lock
  - If lock held, and by a thread running on another CPU, spins
  - If lock held by non-run-state thread, block and sleep waiting for signal of lock being released
- Uses **condition variables**
- Uses **readers-writers** locks when longer sections of code need access to data
- Uses **turnstiles** to order the list of threads waiting to acquire either an adaptive mutex or reader-writer lock
  - Turnstiles are per-lock-holding-thread, not per-object
- Priority-inheritance per-turnstile gives the running thread the highest of the priorities of the threads in its turnstile

# Windows XP Synchronization

- Uses interrupt masks to protect access to global resources on uniprocessor systems
- Uses **spinlocks** on multiprocessor systems
  - Spinlocking-thread will never be preempted
- Also provides **dispatcher objects** user-land which may act mutexes, semaphores, events, and timers
  - **Events**
    - An event acts much like a condition variable
  - Timers notify one or more thread when time expired
  - Dispatcher objects either **signaled-state** (object available) or **non-signaled state** (thread will block)

# Linux Synchronization

- Linux:
  - Prior to kernel Version 2.6, disables interrupts to implement short critical sections
  - Version 2.6 and later, fully preemptive
- Linux provides:
  - semaphores
  - spinlocks
  - reader-writer versions of both
- On single-cpu system, spinlocks replaced by enabling and disabling kernel preemption

# Pthreads Synchronization

- Pthreads API is OS-independent
- It provides:
  - mutex locks
  - condition variables
- Non-portable extensions include:
  - read-write locks
  - spinlocks

# Atomic Transactions

- System Model
- Log-based Recovery
- Checkpoints
- Concurrent Atomic Transactions

# System Model

- Assures that operations happen as a single logical unit of work, in its entirety, or not at all
- Related to field of database systems
- Challenge is assuring atomicity  despite computer system failures
- Transaction - collection of instructions or operations that performs single logical function
  - Here we are concerned with changes to stable storage – disk
  - Transaction is series of read and write operations
  - Terminated by commit  (transaction successful) or abort (transaction failed) operation
  - Aborted transaction must be rolled back to undo any changes it performed

# Types of Storage Media

- Volatile storage – information stored here does not survive system crashes
  - Example: main memory, cache
- Nonvolatile storage – Information usually survives crashes
  - Example: disk and tape
- Stable storage – Information never lost
  - Not actually possible, so approximated via replication or RAID to devices with independent failure modes

Goal is to assure transaction atomicity where failures cause loss of information on volatile storage

# Log-Based Recovery

- Record to stable storage information about all modifications by a transaction
- Most common is write-ahead logging
  - Log on stable storage, each log record describes single transaction write operation, including
    - Transaction name
    - Data item name
    - Old value
    - New value
  - <$T_i$ starts> written to log when transaction $T_i$ starts
  - <$T_i$ commits> written when $T_i$ commits
- Log entry must reach stable storage before operation on data occurs

# Log-Based Recovery Algorithm

- Using the log, system can handle any volatile memory errors
  - Undo($T_i$) restores value of all data updated by $T_i$
  - Redo($T_i$) sets values of all data in transaction $T_i$ to new values
- Undo($T_i$) and redo($T_i$) must be idempotent
  - Multiple executions must have the same result as one execution
- If system fails, restore state of all updated data via log
  - If log contains <$T_i$ starts> without <$T_i$ commits>, undo($T_i$)
  - If log contains <$T_i$ starts> and <$T_i$ commits>, redo($T_i$)

# Checkpoints

- Log could become long, and recovery could take long
- Checkpoints shorten log and recovery time.
- Checkpoint scheme:
  1. Output all log records currently in volatile storage to stable storage
  2. Output all modified data from volatile to stable storage
  3. Output a log record <checkpoint> to the log on stable storage
- Now recovery only includes Ti, such that Ti started executing before the most recent checkpoint, and all transactions after Ti All other transactions already on stable storage

# Concurrent Transactions

- Must be equivalent to serial execution – serializability
- Could perform all transactions in critical section
  - Inefficient, too restrictive
- Concurrency-control algorithms provide serializability

# Serializability

- Consider two data items A and B
- Consider Transactions $T_0$ and $T_1$
- Execute $T_0$, $T_1$ atomically
- Execution sequence called schedule
- Atomically executed transaction order called serial schedule
- For N transactions, there are N! valid serial schedules

# Schedule 1: $T_0$ then $T_1$

| $T_0$ | $T_1$ |
|---|---|
| read($A$) | |
| write($A$) | |
| read($B$) | |
| write($B$) | |
| | read($A$) |
| | write($A$) |
| | read($B$) |
| | write($B$) |

# Nonserial Schedule

- Nonserial schedule allows overlapped execute
  - Resulting execution not necessarily incorrect
- Consider schedule S, operations $O_i$, $O_j$
  - Conflict if access same data item, with at least one write
- If $O_i$, $O_j$ consecutive and operations of different transactions & $O_i$ and $O_j$ don't conflict
  - Then S' with swapped order $O_j$ $O_i$ equivalent to S
- If S can become S' via swapping nonconflicting operations
  - S is conflict serializable

# Schedule 2: Concurrent Serializable Schedule

| $T_0$ | $T_1$ |
|---|---|
| read($A$) | |
| write($A$) | |
| | read($A$) |
| | write($A$) |
| read($B$) | |
| write($B$) | |
| | read($B$) |
| | write($B$) |

# Locking Protocol

- Ensure serializability by associating lock with each data item
  - Follow locking protocol for access control
- Locks
  - Shared – $T_i$ has shared-mode lock (S) on item Q, $T_i$ can read Q but not write Q
  - Exclusive – Ti has exclusive-mode lock (X) on Q, $T_i$ can read and write Q
- Require every transaction on item Q acquire appropriate lock
- If lock already held, new request may have to wait
  - Similar to readers-writers algorithm

# Two-phase Locking Protocol

- Generally ensures conflict serializability
- Each transaction issues lock and unlock requests in two phases
  - Growing – obtaining locks
  - Shrinking – releasing locks
- Does not prevent deadlock

# Timestamp-based Protocols

- Select order among transactions in advance – timestamp-ordering
- Transaction $T_i$ associated with timestamp $TS(T_i)$ before $T_i$ starts
  - $TS(T_i) < TS(T_j)$ if Ti entered system before $T_j$
  - TS can be generated from system clock or as logical counter incremented at each entry of transaction
- Timestamps determine serializability order
  - If $TS(T_i) < TS(T_j)$, system must ensure produced schedule equivalent to serial schedule where $T_i$ appears before $T_j$

# Timestamp-based Protocol Implementation

- Data item Q gets two timestamps
  - W-timestamp(Q) – largest timestamp of any transaction that executed write(Q) successfully
  - R-timestamp(Q) – largest timestamp of successful read(Q)
  - Updated whenever read(Q) or write(Q) executed
- Timestamp-ordering protocol assures any conflicting read and write executed in timestamp order
- Suppose Ti executes read(Q)
  - If $TS(T_i) < \text{W-timestamp}(Q)$, Ti needs to read value of Q that was already overwritten
    - read operation rejected and $T_i$ rolled back
  - If $TS(T_i) \geq \text{W-timestamp}(Q)$
    - read executed, R-timestamp(Q) set to max(R-timestamp(Q), $TS(T_i)$)

# Timestamp-ordering Protocol

- Suppose Ti executes write(Q)
  - If $TS(T_i)$ < R-timestamp(Q), value Q produced by $T_i$ was needed previously and $T_i$ assumed it would never be produced
    - Write operation rejected, $T_i$ rolled back
  - If $TS(T_i)$ < W-timestamp(Q), $T_i$ attempting to write obsolete value of Q
    - Write operation rejected and $T_i$ rolled back
  - Otherwise, write executed
- Any rolled back transaction $T_i$ is assigned new timestamp and restarted
- Algorithm ensures conflict serializability and freedom from deadlock

# Schedule Possible Under Timestamp Protocol

| $T_2$ | $T_3$ |
|---|---|
| read($B$) | |
| | read($B$) |
| | write($B$) |
| read($A$) | |
| | read($A$) |
| | write($A$) |

# EXERCISE

- 5.9
- 5.12
- 5.15
- 5.17
- 5.22

# End of Chapter 6